

# Yingru Li

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*Curriculum Vitae*

## EDUCATION

**The Chinese University of Hong Kong (CUHK)** 2025

Ph.D. in Computer Science and Information Engineering.

Advisor: Zhi-Quan (Tom) Luo.

Committee: Xinyun Chen, Baoxiang Wang, John C.S. Lui, Benjamin Van Roy (Stanford & DeepMind).

**Huazhong University of Science and Technology (HUST)** 2020

B.E./M.S. in Computer Science. Ranked 1st/134 overall; 1st/26 in Software and Theoretical CS.

## PROFESSIONAL EXPERIENCE

**xAI - SpaceX** Singapore  
Member of Technical Staff December 2025 – Present

- Contribute to large-scale (“big-run”) reinforcement learning for the Grok models, spanning both the **science** and the **infrastructure** of RL training.
- **Science of RL** — stability, efficiency, and the core algorithms, recipes, optimizer and scaling laws; helped identify and resolve critical training instabilities that benefit every stage of the RL pipeline, contributing to one of the longest stable RL runs to date.
- **Long-horizon RL** — working on training methods and recipes for long-horizon agentic reinforcement learning, including higher intelligence per token and credit assignment.
- **RL infrastructure** — helped build and scale the distributed training systems behind these recipes.

**ByteDance** Singapore  
LLM Algorithm Research Scientist 2025 – December 2025

- Led research on RL training stability and efficiency for LLM reasoning and agents.
- Diagnosed and formalized the **RL Collapse** phenomenon, tracing instability to implicit off-policy caused by the training–inference mismatch; developed Trust Region Masking, Dynamic Vocabulary Pruning, the Optimal Token Baseline, and a simple LR-scheduling fix.
- Co-led SimpleTIR (end-to-end multi-turn tool-integrated reasoning) and Dr. Kernel (Triton kernel code generation agent via RL).

**Tencent AI & Robotics X** Shenzhen, China  
Student Research Staff Member, Agent Center 2019 – 2022

- Led HyperAgent (ICML 2024) and HyperDQN (ICLR 2022) for scalable exploration in deep RL, and Divergence-Augmented Policy Optimization (NeurIPS 2019) for stable off-policy data reuse.

**Microsoft Research** Beijing, China  
Research Intern, Theory Center 2016

- Research on combinatorial optimization and reinforcement / bandit learning algorithms.

## PUBLICATIONS

Full list at [Google Scholar](#). ★ denotes equal contribution; † denotes corresponding author.

### Peer-Reviewed Conference & Journal Publications

1. **The Optimal Token Baseline: Variance Reduction for Long-Horizon LLM-RL**  
Yingru Li, Jiawei Xu, Ziniu Li, Jiakai Liu, Wei Liu, Yuxuan Tong, Longtao Zheng, Zhenghai Xue, Yaxiang Zhang, Tianle Cai, Ge Zhang, Qian Liu, Baoxiang Wang. *ICML 2026*.
2. **Trust Region Masking for Long-Horizon LLM Reinforcement Learning**  
Yingru Li, Jiakai Liu, Jiawei Xu, Yuxuan Tong, Ziniu Li, Qian Liu, Baoxiang Wang. *ICML 2026*.

3. **Optimistic Thompson Sampling for No-Regret Learning in Unknown Games**  
Yingru Li, Liangqi Liu, Wenqiang Pu, Hao Liang, Zhi-Quan Luo. *IEEE Transactions on Signal Processing*, 2026.
4. **Dr. Kernel: Reinforcement Learning Done Right for Triton Kernel Generations**  
Wei Liu, Jiawei Xu, Yingru Li, Longtao Zheng, Tianjian Li, Qian Liu, Junxian He. *ICML 2026*.
5. **Harnessing Uncertainty: Entropy-Modulated Policy Gradients for Long-Horizon LLM Agents**  
Jiawei Wang, Jiakai Liu, Yuqian Fu, Yingru Li, Xintao Wang, Yuan Lin, Yu Yue, Lin Zhang, Yang Wang, Ke Wang. *ICML 2026*.
6. **SimpleTIR: End-to-End Reinforcement Learning for Multi-Turn Tool-Integrated Reasoning**  
Zhenghai Xue, Longtao Zheng, Qian Liu, Yingru Li, Xiaosen Zheng, Zejun Ma, Bo An. *ICLR 2026*.
7. **Scalable Exploration via Ensemble++**  
Yingru Li, Jiawei Xu, Baoxiang Wang, Zhi-Quan Luo. *NeurIPS 2025*.
8. **Owl: Optimized Workforce Learning for General Multi-Agent Assistance in Real-World Task Automation**  
Mengkan Hu, Yuhang Zhou, Wendong Fan, Yuzhou Nie, Bowei Xia, Tao Sun, Ziyu Ye, Zhaoxuan Jin, Yingru Li, Qiguang Chen, Zeyu Zhang, Yifeng Wang, Qianshuo Ye, Bernard Ghanem, Ping Luo, Guohao Li. *NeurIPS 2025*.
9. **Q-Star Meets Scalable Posterior Sampling: Bridging Theory and Practice via HyperAgent**  
Yingru Li, Jiawei Xu, Lei Han, Zhi-Quan Luo. *ICML 2024*.
10. **Prior-Dependent Analysis of Posterior Sampling RL with Function Approximation**  
Yingru Li, Zhi-Quan Luo. *AISTATS 2024*.
11. **HyperDQN: A Randomized Exploration Method for Deep Reinforcement Learning**  
Ziniu Li, Yingru Li<sup>†</sup>, Yushun Zhang, Tong Zhang, Zhi-Quan Luo. *ICLR 2022*.
12. **Divergence-Augmented Policy Optimization**  
Qing Wang\*, Yingru Li\*, Jiechao Xiong, Tong Zhang. *NeurIPS 2019*.

## Preprints

1. **When Speed Kills Stability: Demystifying RL Collapse from the Training-Inference Mismatch**  
Jiakai Liu\*, Yingru Li<sup>\*,†</sup>, Yuqian Fu, Jiawei Wang, Qian Liu, Yu Shen. *Technical Blog*, 2025.
2. **Beyond Precision: Training-Inference Mismatch is an Optimization Problem and Simple LR Scheduling Fixes It**  
Yaxiang Zhang\*, Yingru Li<sup>\*,†</sup>, Jiakai Liu, Jiawei Xu, Ziniu Li, Qian Liu, Haoyuan Li. *arXiv 2026*.
3. **Learning an Opponent-Aware Anti-Jamming Strategy via Online Convex Optimization**  
Liangqi Liu, Wenqiang Pu, Yingru Li, Zhi-Quan Luo. *Under review (minor revision), IEEE Transactions on Signal Processing; arXiv 2026*.
4. **Taming the Tail: Stable LLM Reinforcement Learning via Dynamic Vocabulary Pruning**  
Yingru Li, Jiawei Xu, Jiakai Liu, Yuxuan Tong, Ziniu Li, Tianle Cai, Ge Zhang, Qian Liu, Baoxiang Wang. *arXiv 2025*.
5. **A Note on Hybrid Online RL and Imitation Learning for LLMs: Formulations and Algorithms**  
Yingru Li, Ziniu Li, Jiakai Liu. *arXiv 2025*.
6. **Logit Dynamics in Softmax Policy Gradient Methods**  
Yingru Li. *arXiv 2025*.
7. **Scaling Flaws of Verifier-Guided Search in Mathematical Reasoning**  
Fei Yu, Yingru Li, Benyou Wang. *arXiv 2025*.
8. **Uncertainty-Aware Search and Value Models: Mitigating Search Scaling Flaws in LLMs**  
Fei Yu, Yingru Li, Benyou Wang. *arXiv 2025*.
9. **Probability Tools for Sequential Random Projection**  
Yingru Li. *arXiv 2024*.
10. **Simple, Unified Analysis of Johnson-Lindenstrauss with Applications**  
Yingru Li. *arXiv 2024*.

## PATENTS

- Liangqi Liu, Wenqiang Pu, **Yingru Li**, Zhi-Quan Luo. *Game-Strategy Update Method, Device, and Medium for Unknown Matrix Games*. Chinese invention patent application, CN117787419A (Shenzhen Research Institute of Big Data; published Mar. 2024).

## RESEARCH IMPACT & ADOPTION

The training–inference mismatch / RL-collapse line of work (*When Speed Kills Stability*) has been adopted across major open-source RL training stacks and frontier models:

- **DeepSeek-V3.2** adopts the geometric sequence masking to address the training–inference mismatch and off-policy training instability ([arXiv:2512.02556](#)).
- Integrated into **veRL** (v0.6.0+): geometric-mean importance-sampling masking / rejection (Geo-MIS / Geo-RS) and **rollout correction** for general off-policy training; Trust Region Masking and the Optimal Token Baseline are also available in veRL.
- Integrated into **Slime** and **Miles**. Discussed in a **vLLM engineering blog** (bitwise-consistent on-policy RL) and a **SGLang Blog**.
- Used by **Cognition’s SWE-grep** (sequence-level masked importance sampling) and discussed by the **Qwen team**.

## AWARDS & HONORS

- **Best Paper Award**, 3rd Doctoral and Postdoctoral Daoyuan Academic Forum, 2024.
- **Best Student Paper**, IEEE Sensor Array and Multichannel Signal Processing Workshop (SAM), 2024.
- **SRIBD Ph.D. Fellowship (Gold Class)**, Shenzhen Research Institute of Big Data, 2023.
- **Presidential Ph.D. Fellowship**, The Chinese University of Hong Kong, 2019–2023.
- **Tencent AI Ph.D. Fellowship**, Tencent & The Chinese University of Hong Kong, 2018.
- **National Scholarship (China)** — highest national academic honor, top 0.2% nationwide, 2018.
- **Qiming Star Award** — 1 of 5 recipients from 7,112 undergraduates, HUST, 2016.
- **First Prize**, Parallel Computation and Application Contest (PAC), Intel & CCF, 2015.
- **First Prize**, China National Mathematics Olympiad (Province-level Math League), 2012.

## PEER REVIEW & PROFESSIONAL SERVICE

Reviewer for the leading peer-reviewed venues in machine learning and artificial intelligence. NeurIPS, ICML, and ICLR are the three top-tier conferences in the field; AISTATS is a leading venue in machine-learning theory and statistics; TMLR is a flagship ML journal.

| Venue                      | Role                  | Papers reviewed  |
|----------------------------|-----------------------|------------------|
| NeurIPS (2023, 2024, 2025) | Reviewer              | 6 + 6 + 5 = 17   |
| ICLR (2024, 2025, 2026)    | Reviewer              | 2 + 3 + 5 = 10   |
| ICML (2025, 2026)          | Reviewer <sup>†</sup> | 5 + 6 = 11       |
| AISTATS (2025)             | Reviewer              | 2                |
| TMLR (ongoing)             | Reviewer              | 3                |
| <b>Total (2023–2026)</b>   |                       | <b>43 papers</b> |

- † **ICML 2026 Gold Reviewer** — recognized by the ICML Program Chairs as among the top reviewers for the conference, based on area-chair evaluations of the quality of reviews submitted; awarded complimentary registration.
- Admitted to the NeurIPS 2026 and AISTATS 2026 reviewer pools (paper assignments pending).
  - **28 additional papers** reviewed across 9 affiliated workshops, including ICLR Workshop “Bridging the Gap Between Practice and Theory in Deep Learning,” ICML Workshop “Aligning RL Experimentalists and The-

orists,” and NeurIPS Workshops on Bayesian Decision-making & Uncertainty and RL4RealLife.

- **Session Chair**, 2 sessions on “Integrating Generative AI with Sequential Decision-making,” INFORMS Annual Meeting 2024.
- **Organizer / Chair**, Reinforcement Learning Seminar, CUHK (8 semesters, 2019–2022).

#### INVITED TALKS & ORAL PRESENTATIONS

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- **Revisiting Trust Region Policy Optimization in the Era of LLMs** — Invited talk, QingkeAI (Dec. 2025).
- **LLM-RL Instability and the Training–Inference Mismatch** — Invited talks at SLAI and the Bay Area RL community (Nov. 2025).
- **Tackling Data Scarcity for Trustworthy Agents** — Invited talk, ETH Zurich (Nov. 2024).
- **Exploration at Scale: Theory, Algorithms & Applications** (*a.k.a. Scalable Uncertainty Quantification for Exploration and LLM Reasoning*) — Invited talks at MIT (Jul. 2024), Princeton University (May 2024), INFORMS Annual Meeting (Seattle, Oct. 2024), International Symposium on Mathematical Programming / ISMP (Montréal, Jul. 2024), INFORMS Optimization Society Conference at Rice University (Mar. 2024), and RLChina (Jun. 2024); contributed talk at the 3rd Daoyuan Academic Forum (Jan. 2024).
- **No-Regret Learning in Unknown Games with Applications** — Invited talk, RL Theory Student Workshop, Nanjing University (Aug. 2022); contributed talk, 2nd Daoyuan Academic Forum (Aug. 2022).
- **HyperDQN: A Randomized Exploration Method for Deep RL** — Contributed talk, NeurIPS Workshop on Ecological Theory of RL (Dec. 2021).

#### SELECTED MEDIA COVERAGE

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- Featured for the Qiming Star Award in the HUST campus newspaper and HUST Online, 2016.

#### TEACHING

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Teaching Assistant, The Chinese University of Hong Kong (Shenzhen) — duties delivered in English (tutorials, grading, laboratory sessions): Stochastic Processes (Fall 2018), Optimization II (Spring 2019), Distributed and Parallel Computing (Fall 2019), Reinforcement Learning (Fall 2020), Matrix Analysis (Spring 2021), and Deep Learning and Their Applications (Spring 2022).